Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

END SEMESTER REVIEW

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REACT JS BASED WEBAPPLICATION FOR TICKET BOOKING OF AN INTERNAL DEPARTMENT EVENT

SDG: SDG No – SDG 9 – Industry, Innovation and Infrastructure

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INTRODUCTION

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- The project focuses on building a Smart Ticket Booking System.
- It allows users to book tickets for events easily.
- The system automates booking, payment simulation, and receipt generation.
- Users receive a downloadable PDF receipt after booking.
- Reduces manual effort and improves efficiency.
- Useful for colleges, seminars, and events.
- Built using Full Stack Development concepts
- Admin panel allows various operations like edit, delete and update events

PROBLEM STATEMENT

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- Manual ticket booking systems are slow and inefficient.
- Data handling errors are common in traditional systems.
- No proper tracking of bookings and receipts.
- Lack of automation leads to poor user experience.
- Need for a digital system to manage bookings efficiently.
- Requirement for instant confirmation and receipt generation
- Existing systems do not provide automatic email confirmation or digital receipts, making it difficult for users to track bookings.
- Lack of centralized management for events and bookings, causing difficulty in monitoring availability and user activity.

OBJECTIVE(S)

- To design and develop a full-stack ticket booking system.
- To implement automated booking, payment, and admin panel.
- To provide user friendly interface and glitch free database connection
- To provide CRUD operations by admin
- To provide seat booking by students
- To provide the history of tickets booked by them
- Enabling credit card based payment

PROPOSED SYSTEM

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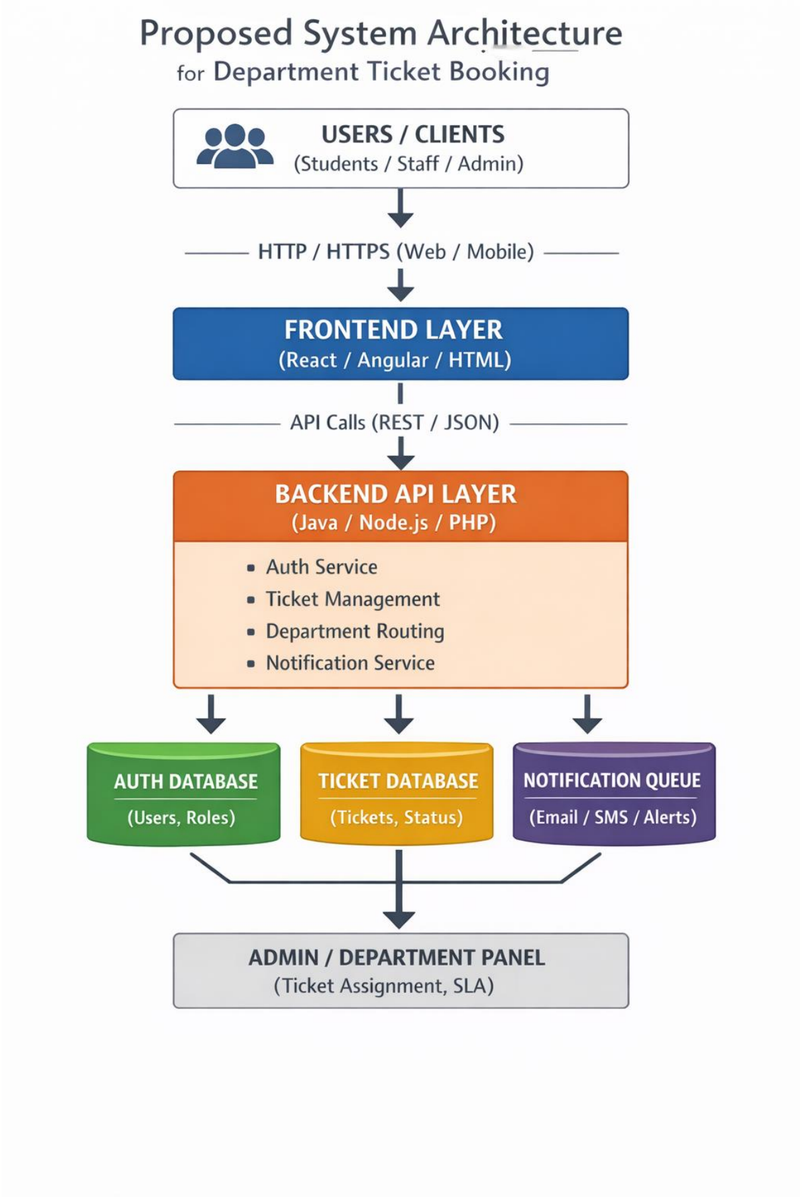
- Online ticket booking interface for users. Secure login system (User & Admin).
- Event selection and ticket booking module.After login, users can view event details.
- Users can select number of tickets to book.It checks email format and required fields.Admin panel for managing events. Booking history tracking system
- It checks email format and required fields.
- Admin panel for managing events
- Displays seats total count, dynamic updation of booked seats.
- Display history of tickets booked by the student

PROPOSED SYSTEM ARCHITECTURE



PROPOSED SYSTEM ARCHITECTURE

- **Input→**
  User enters details (name, email, tickets)
- **Processing→**
  Backend handles booking + payment logic
- **Database→**
  Stores users, events, bookings
- **Output→**
  Registration confirmation

WORKFLOW

Workflow
[Start]
[User Login]
[Select Event]
[Enter Booking Details]
[Proceed to Payment]
[Confirm Booking]
[Backend Processing]
[Store Data in Database]
[Notification Given]
[Database Stored]
[End]

TECHNOLOGIES USED

- **Frontend:** React.js, HTML, CSS
- **Backend:** Node.js, Express.js
- **Database:** MySQL , php ,xampp server
- **Tools:** VS Code, Postman
- **Ai**: Antigravity, Claude, Chatgpt

MODULE DESCRIPTION

➢ Module 1:User Module

- Handles secure user login and authentication
- Allows users to select events and book tickets
- Enables users to enter personal and booking details
- Provides payment simulation and booking confirmation
- Displays booking history with option to download PDF receipts

➢ Module 2:Admin Module

- Allows admin to create new events with details (name, date, venue, price)
- Enables updating/editing existing event information
- Provides option to delete events from the system
- Manages ticket availability and updates remaining seats
- Displays overview of all events and booking statistics

MODULE DESCRIPTION

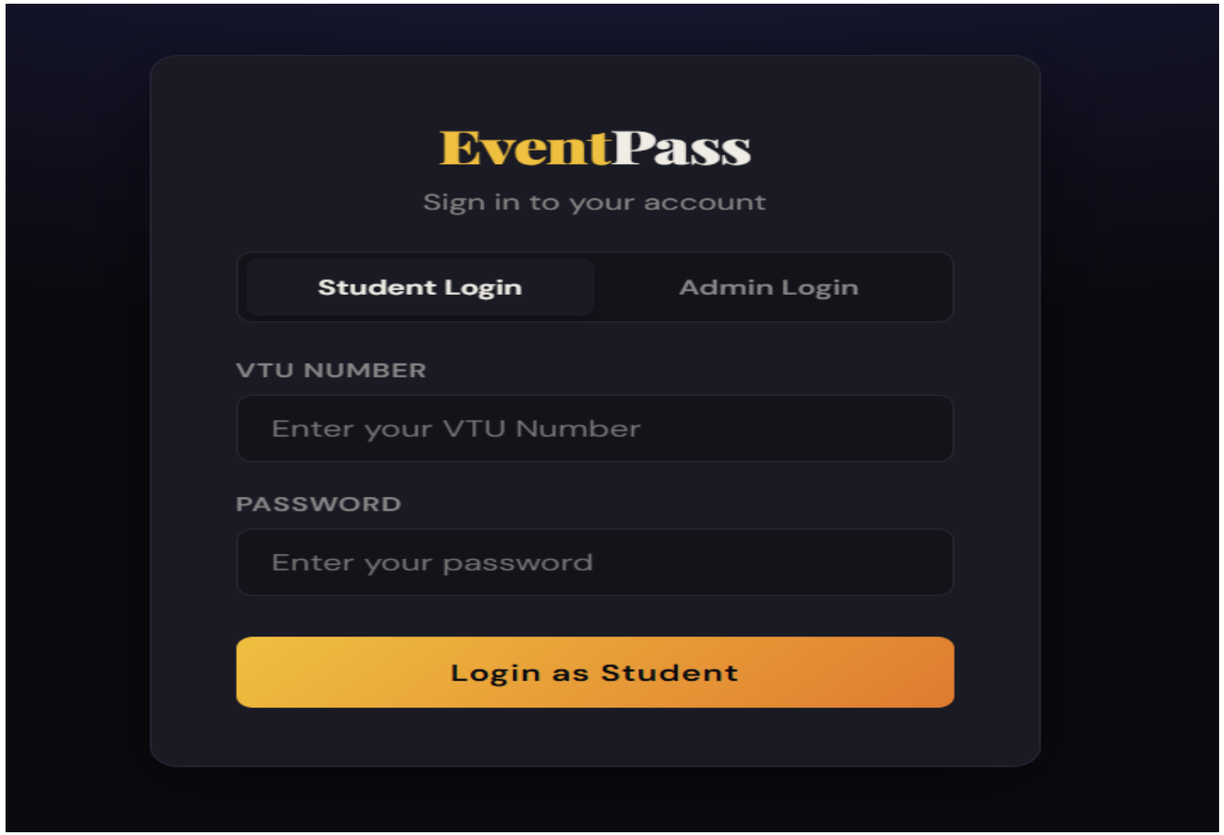
➢ Module3:Booking Module

- Allows users to select events and number of tickets
- Collects user details like name, email, and department
- Calculates total ticket amount based on selection
- Validates ticket availability before confirming booking
- Stores booking details in database for future reference

➢ Module 4:Payment Module

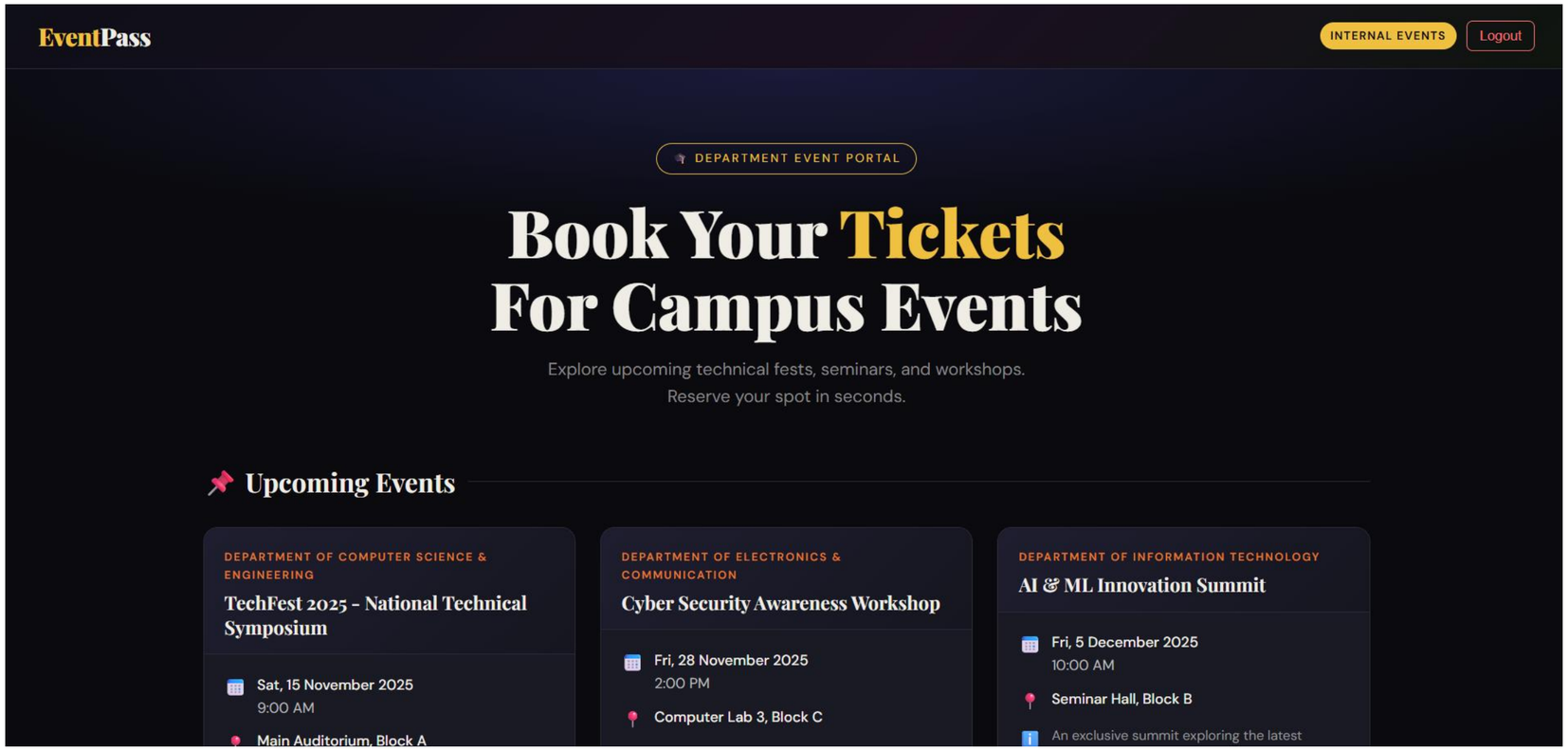
- Simulates the payment process for ticket booking
- Displays total amount based on selected tickets
- Allows user to confirm payment action
- Generates payment success status after confirmation
- Triggers booking confirmation, PDF receipt, and email notification

IMPLEMENTATION SCREENSHOTS



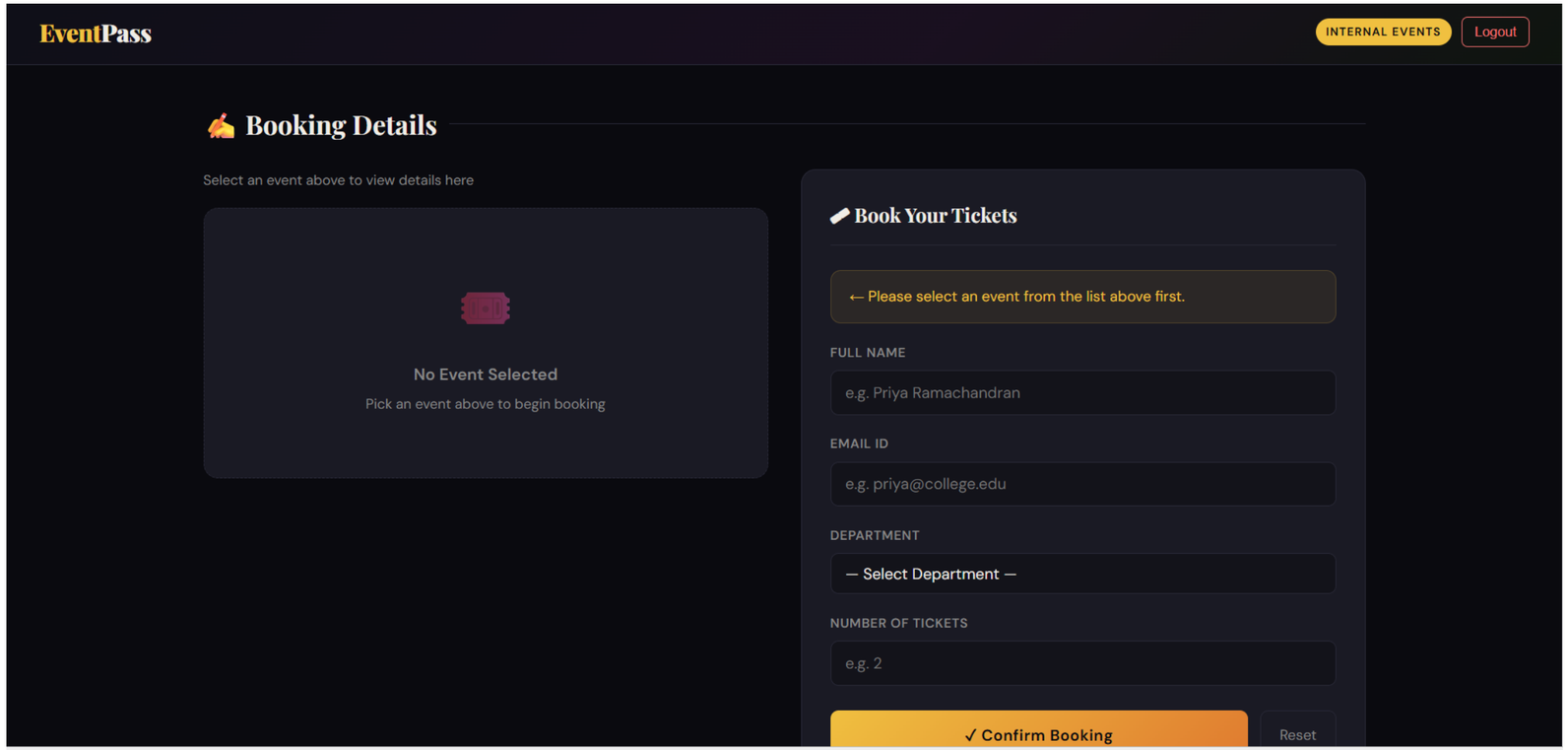
- Login page:
  - Provides secure login for users and admins using email and password
  - Validates credentials and redirects users to respective dashboards

IMPLEMENTATION SCREENSHOTS



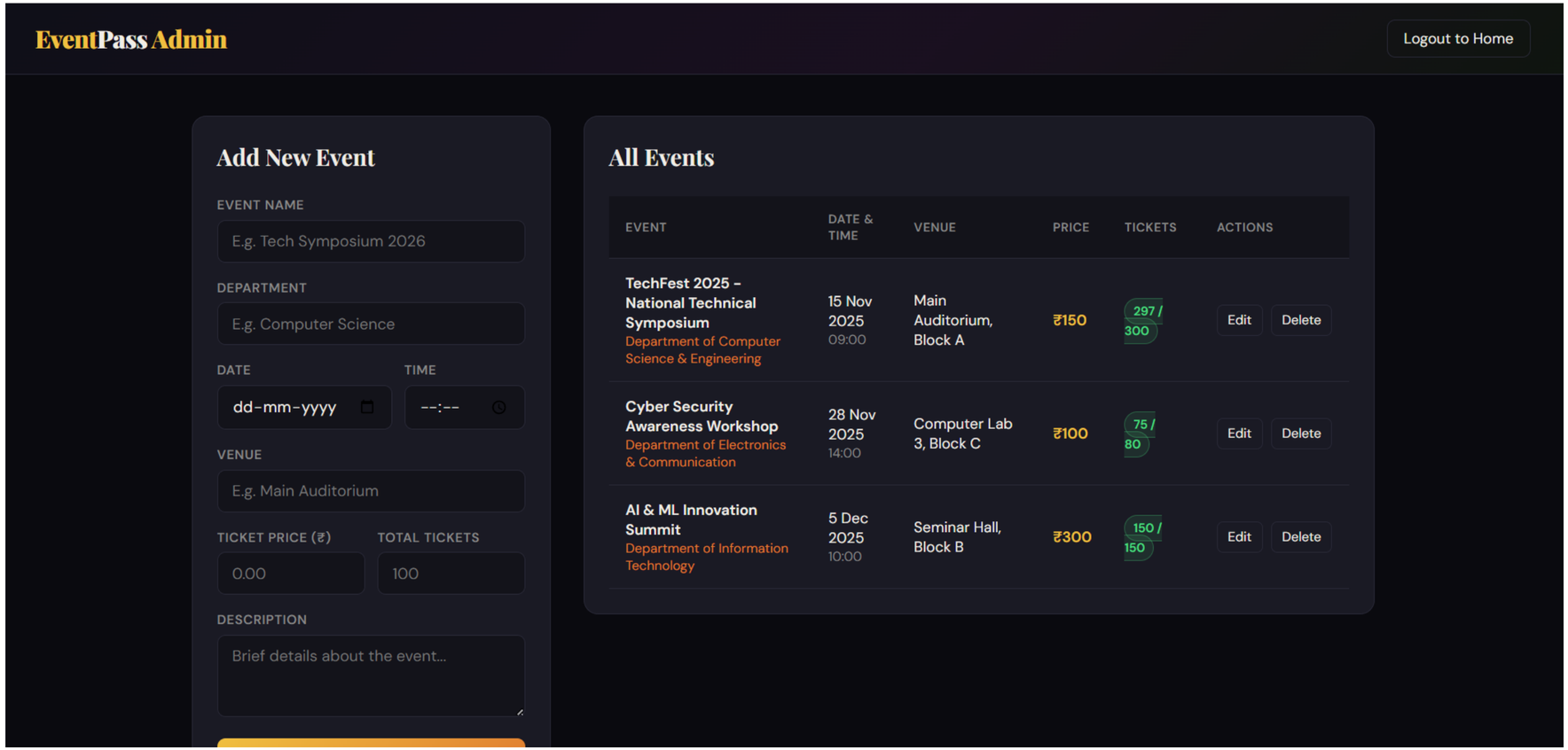
- Booking Page:
  - Allows users to select event, enter details, and choose number of tickets
  - Displays ticket availability and calculates total amount dynamically

IMPLEMENTATION SCREENSHOTS



- Booking Page:
  - Displays booking summary and allows user to enter payment details (card info)
  - Simulates payment process and confirms booking on successful payment

IMPLEMENTATION SCREENSHOTS



- Admin Page:
  - Allows admin to create, update, and delete event details
  - Provides overview of events and manages ticket availability

ADVANTAGES & LIMITATIONS

- **Advantages:**
  - Easy to use
  - Fully automated
  - Saves time
  - Instant ticket generation
  - Email confirmation
- **Limitations:**
  - Payment is simulated (not real gateway)
  - Requires internet
  - Limited to predefined events
  - Not a mobile application

Thank you